Lecture 9: Compatible Projections

MATH203 Applied Linear Algebra

Review: What Cross-Filling Discovered

Recall from Lecture 8

We cross-filled a projection $P = P^2$:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_2}$$

We discovered two **remarkable** properties:

1. Each R_i is a projection: $R_1^2 = R_1$, $R_2^2 = R_2$
2. Mutual annihilation: $R_1 R_2 = 0$, $R_2 R_1 = 0$

The Two Properties Together

Property 1 alone is not special — many matrices satisfy $R^2 = R$.

Property 2 alone is not special — many pairs satisfy $AB = 0$.

But **together** they form a remarkable structure:

- Each R_i projects onto its own "floor"
- $R_i R_j = 0$ means: the floor of R_j lies entirely in the **sunlight direction** of R_i

Today: We name this structure, prove a stunning theorem about it, and discover powerful decomposition methods.

Compatible Families

Definition

Definition (Compatible Family)

A collection of projections $\{P_1, P_2, \ldots, P_k\}$ (each satisfying $P_i^2 = P_i$) is called a **compatible family** if:

$$P_i P_j = 0 \quad \text{for all } i \neq j$$

Example: Coordinate Projections in $\mathbb{R}^3$

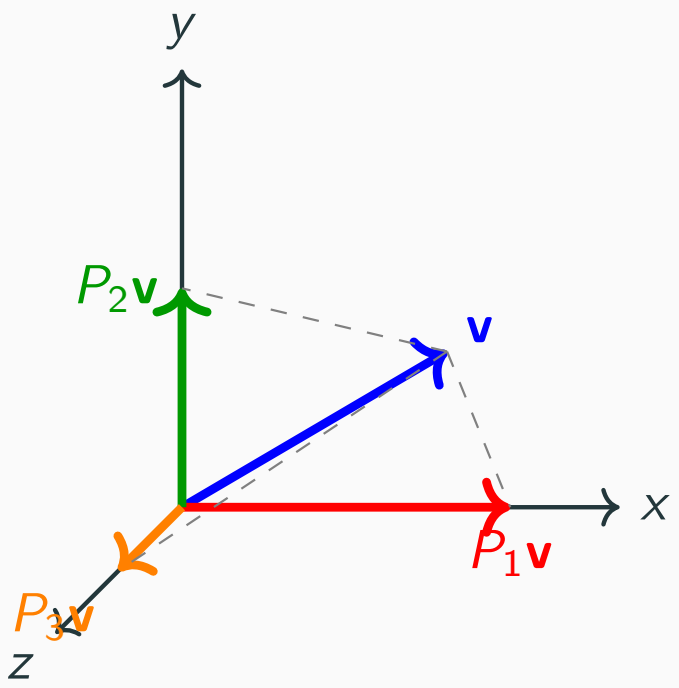
$$P_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Check: each is a projection ($P_i^2 = P_i$ ✓)

Check: $P_1 P_2 = 0$, $P_1 P_3 = 0$, $P_2 P_3 = 0$ ✓

Bonus: $P_1 + P_2 + P_3 = I_3$

Geometric Picture: Coordinate Projections



$$\mathbf{v} = P_1\mathbf{v} + P_2\mathbf{v} + P_3\mathbf{v}$$

Each projection captures one coordinate. The three "floors" (axes) are completely independent.

Example: From Lecture 8's Cross-Filling

Recall:

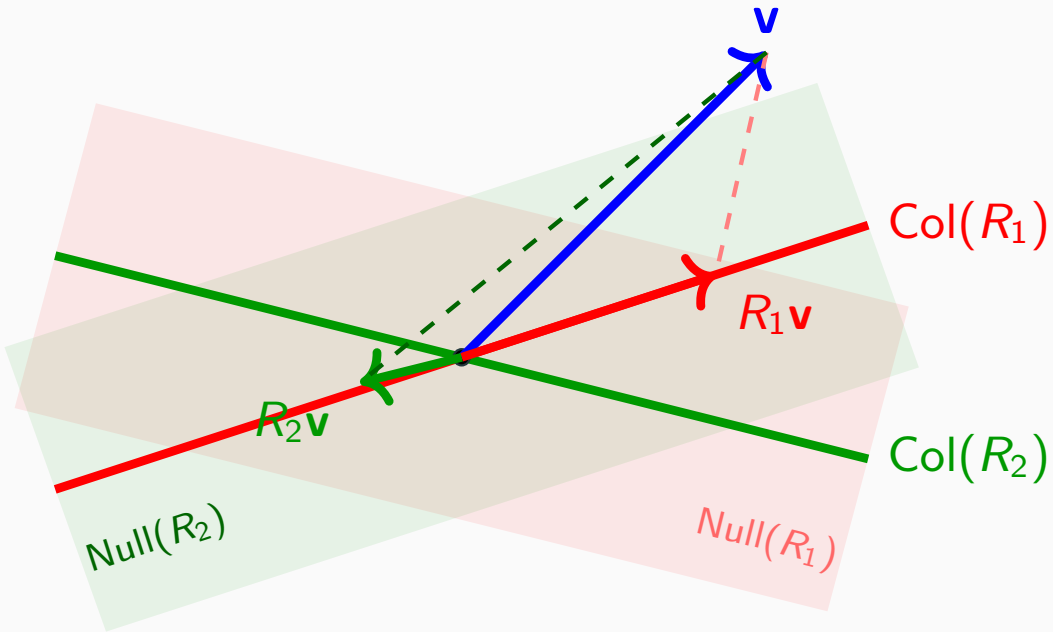
$$R_1 = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

We already verified: $R_1^2 = R_1$, $R_2^2 = R_2$, $R_1 R_2 = 0$.

So $\{R_1, R_2\}$ is a **compatible family**.

This is not a coincidence — we will prove that cross-filling **always** produces compatible families.

Geometric Picture: Non-Orthogonal Projections in Action



Containment (the geometric meaning of $R_1 R_2 = 0$):

- $\mathrm{Col}(R_2) \subset \mathrm{Null}(R_1)$: green line lies inside red plane
- $\mathrm{Col}(R_1) \subset \mathrm{Null}(R_2)$: red line lies inside green plane

Property 1: Subsums Are Projections

Proposition

Let $\{P_1, \ldots, P_k\}$ be a compatible family. Then any subsum

$$Q = P_1 + P_2 + \cdots + P_j$$

is also a projection ($Q^2 = Q$).

Proof: Expand Q^2:

$$Q^2 = (P_1 + P_2 + \cdots + P_j)(P_1 + P_2 + \cdots + P_j)$$

$$= P_1^2 + \underbrace{P_1 P_2}_{=0} + \cdots + P_2^2 + \underbrace{P_2 P_1}_{=0} + \cdots + P_j^2$$

All cross terms vanish ($P_i P_l = 0$), only diagonal terms survive:

$$Q^2 = P_1^2 + P_2^2 + \cdots + P_j^2 = P_1 + P_2 + \cdots + P_j = Q \quad \checkmark$$

Verify with Coordinate Projections

$$P_1 + P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ is a projection } \checkmark \qquad \text{(projects onto } xy\text{-plane)}$$

$$P_2 + P_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is a projection } \checkmark \qquad \text{(projects onto } yz\text{-plane)}$$

$$P_1 + P_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is a projection } \checkmark \qquad \text{(projects onto } xz\text{-plane)}$$

$P_1 + P_2 + P_3 = I_3$ is a projection $\checkmark$

Property 2: Independent Decompositions

Proposition

Let $\{P_1, \ldots, P_k\}$ be a compatible family. For any vector $\mathbf{v}$, the nonzero vectors among $\{P_1\mathbf{v}, \ldots, P_k\mathbf{v}\}$ are linearly independent.

Proof: Suppose $c_1 P_1 \mathbf{v} + \cdots + c_k P_k \mathbf{v} = \mathbf{0}$.

Left-multiply by P_j:

$$c_1 \underbrace{P_j P_1}_{=0} \mathbf{v} + \cdots + c_j \underbrace{P_j^2}_{=P_j} \mathbf{v} + \cdots + c_k \underbrace{P_j P_k}_{=0} \mathbf{v} = \mathbf{0}$$

$$c_j P_j \mathbf{v} = \mathbf{0}$$

If $P_j \mathbf{v} \neq \mathbf{0}$, then $c_j = 0$. $\square$

Unique Decomposition

If $\{P_1, \ldots, P_k\}$ is a compatible family with $P_1 + \cdots + P_k = P$, then every vector $P\mathbf{v}$ decomposes as:

$$P\mathbf{v} = \underbrace{P_1\mathbf{v}}_{\in \mathrm{Col}(P_1)} + \underbrace{P_2\mathbf{v}}_{\in \mathrm{Col}(P_2)} + \cdots + \underbrace{P_k\mathbf{v}}_{\in \mathrm{Col}(P_k)}$$

By Property 2 (linear independence), this decomposition is **unique**.

(In other textbooks, this is called a **direct sum decomposition**, written $\mathrm{Col}(P) = \mathrm{Col}(P_1) \oplus \cdots \oplus \mathrm{Col}(P_k)$. We will not use this notation.)

Key: $P_i P_j = 0$ ensures the "floors" are independent — the floor of P_i lies in the sunlight direction of P_j.

The Projection Decomposition Theorem

The Big Question

Compatible families seem like a very **special** structure. We required:

1. Each $P_i^2 = P_i$ (projection)
2. $P_i P_j = 0$ for $i \neq j$ (mutual annihilation)

Suppose we write a projection as a sum:

$$P = R_1 + R_2 + \cdots + R_k$$

We don't assume the R_i are projections. We don't assume $R_i R_j = 0$.

Question: Under what conditions can we **guarantee** that $\{R_1, \ldots, R_k\}$ is a compatible family?

The Stunning Answer

Theorem (Projection Decomposition — The Climax)

Let P be a projection ($P^2 = P$). Suppose:

$$P = R_1 + R_2 + \cdots + R_k$$

where each R_i is a matrix (not necessarily a projection!), and:

$$\text{rank}(R_1) + \text{rank}(R_2) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$$

Then **automatically**:

1. Each R_i is a projection ($R_i^2 = R_i$)
2. $R_i R_j = 0$ for all $i \neq j$ (compatible family)
3. Equality holds: $\text{rank}(R_1) + \cdots + \text{rank}(R_k) = \text{rank}(P)$

Why Is This Remarkable?

We **never assumed** the R_i are projections.

We **never assumed** $R_i R_j = 0$.

The projection property and mutual annihilation **emerge automatically** from the rank condition alone!

To check compatibility, we only need:

$$\text{rank}(R_1) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$$

Key: The rank condition forces a rigid algebraic structure.

Example: Let's Try It

Take a rank-2 projection (from Lecture 8's diagonal cross-filling):

$$P = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \qquad \text{trace} = \frac{2+2+2}{3} = 2 = \text{rank}(P)$$

Cross-fill at diagonal pivot $(1,1)$, then pivot $(2,2)$:

$$P = \underbrace{\frac{1}{6}\begin{pmatrix} \mathbf{4} & -2 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & 1 \end{pmatrix}}_{R_1,\ \text{rank}=1} + \underbrace{\frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{1} & -1 \\ 0 & -1 & 1 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

Red = pivot, blue = cross-arm, black = determined by cross

Example: Check the Rank Condition

$$\text{rank}(R_1) + \text{rank}(R_2) = 1 + 1 = 2 = \text{rank}(P) \quad \checkmark$$

The theorem **predicts** (without any computation!):

1. R_1 is a projection (${R_1}^2 = R_1$)
2. R_2 is a projection (${R_2}^2 = R_2$)
3. $R_1 R_2 = 0$ (compatible)

Quick sanity check: $\text{trace}(R_1) = \frac{4+1+1}{6} = 1, \quad \text{trace}(R_2) = \frac{0+1+1}{2} = 1$

Each piece has trace $= 1$. This is consistent with being a rank-1 projection (since trace $=$ rank for projections).

Corollary: The Trace Test

Corollary (Trace Test)

If $P = R_1 + R_2 + \cdots + R_k$ satisfies the rank condition, then each R_i is a projection, so:

$$\text{trace}(R_i) = \text{rank}(R_i)$$

In particular, if each R_i has rank 1, then each must have $\text{trace}(R_i) = 1$.

Contrapositive: If you decompose $P = R_1 + R_2$ and find $\text{trace}(R_1) \neq \text{rank}(R_1)$, then the rank condition **must fail** — no need to check ranks!

Trace is just reading off the diagonal — much cheaper than computing rank.

Example: Trace Test Catches a Bad Decomposition

Same $P = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$, but decompose differently:

$$P = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{A,\ \text{trace}=2} + \underbrace{\begin{pmatrix} -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}}_{B,\ \text{trace}\approx 0}$$

$\text{trace}(A) = 2$ but $\text{rank}(A) = 2$. Trace $\neq$ rank? That's fine for A itself.

But $\text{rank}(A) + \text{rank}(B) = 2 + 2 = 4 > 2 = \text{rank}(P)$.

Rank condition **fails**. And indeed B is **not** a projection ($B^2 \neq B$).

Proof: Step 1 — Introduce R_0

Define $R_0 = I_n - P$. Since P is a projection, so is R_0:

$$(I_n - P)^2 = I_n - 2P + P^2 = I_n - 2P + P = I_n - P = R_0 \quad \checkmark$$

Now add R_0 to both sides of $P = R_1 + R_2 + \cdots + R_k$:

$$\underbrace{R_0}_{I_n - P} + \underbrace{R_1 + R_2 + \cdots + R_k}_{=P} = I_n$$

We have reduced to the case where the pieces sum to I_n.

Proof: Step 2 — Count Total Rank

Since R_0 is a projection, $\text{rank}(R_0) = \text{trace}(R_0)$ (Lecture 8).

$$\text{trace}(R_0) = \text{trace}(I_n - P) = \underbrace{\text{trace}(I_n)}_{=n} - \text{trace}(P) = n - \text{rank}(P)$$

Now count the total rank of **all** pieces $(R_0, R_1, \ldots, R_k)$:

$$\underbrace{\text{rank}(R_0)}_{=n-\text{rank}(P)} + \underbrace{\text{rank}(R_1) + \cdots + \text{rank}(R_k)}_{\leq \text{rank}(P) \text{ (given)}} \leq n$$

Proof: Step 3 — Factor Each R_i

Write $r_i = \text{rank}(R_i)$. By cross-filling (Lecture 8), each R_i factors as:

$$R_i = \underbrace{U_i}_{n \times r_i} \cdot \underbrace{V_i}_{r_i \times n}$$

Let $s = r_0 + r_1 + \cdots + r_k =$ **total number of columns across all U_i**.

From Step 2: $s \leq n$.

Proof: Step 3 (continued) — Assemble Block Matrices

Stack all U_i side by side, all V_i on top of each other:

$$\underbrace{\begin{pmatrix} | & | & & | \\ U_0 & U_1 & \cdots & U_k \\ | & | & & | \end{pmatrix}}_{U,\ \text{size } n\times s} \quad \underbrace{\begin{pmatrix} - & V_0 & - \\ - & V_1 & - \\ & \vdots & \\ - & V_k & - \end{pmatrix}}_{V,\ \text{size } s\times n}$$

Then:

$$UV = U_0 V_0 + U_1 V_1 + \cdots + U_k V_k = R_0 + R_1 + \cdots + R_k = I_n$$

Proof: Step 4 — $UV = I_n$ Forces $s = n$

We have: $\underbrace{U}_{n \times s} \cdot \underbrace{V}_{s \times n} = I_n$ and $s \leq n$.

Can $s < n$? No! If $s < n$:

$$\text{rank}(UV) \leq \text{rank}(U) \leq s < n$$

But $\text{rank}(I_n) = n$. Contradiction!

So $s = n$, meaning U and V are both **square** ($n \times n$).

Since $UV = I_n$ and both are square: $VU = I_n$ as well (Lecture 5).

Proof: Step 5 — What Does $VU = I_n$ Look Like?

Recall the block structure (from Step 3):

$$\underbrace{\begin{pmatrix} - & V_0 & - \\ - & V_1 & - \\ & \vdots & \\ - & V_k & - \end{pmatrix}}_{V} \cdot \underbrace{\begin{pmatrix} | & | & & | \\ U_0 & U_1 & \cdots & U_k \\ | & | & & | \end{pmatrix}}_{U} = I_n$$

Row i of V times column j of U gives block (i, j):

$$VU = \begin{pmatrix} V_0 U_0 & V_0 U_1 & \cdots & V_0 U_k \\ V_1 U_0 & V_1 U_1 & \cdots & V_1 U_k \\ \vdots & \vdots & \ddots & \vdots \\ V_k U_0 & V_k U_1 & \cdots & V_k U_k \end{pmatrix}$$

Proof: Step 5 (continued) — Reading Off the Blocks

$$\begin{pmatrix} V_0 U_0 & V_0 U_1 & \cdots & V_0 U_k \\ V_1 U_0 & V_1 U_1 & \cdots & V_1 U_k \\ \vdots & \vdots & \ddots & \vdots \\ V_k U_0 & V_k U_1 & \cdots & V_k U_k \end{pmatrix} = \begin{pmatrix} I_{r_0} & 0 & \cdots & 0 \\ 0 & I_{r_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I_{r_k} \end{pmatrix}$$

Diagonal blocks: $V_i U_i = I_{r_i}$ (same color $\to$ identity)

Off-diagonal blocks: $V_i U_j = 0$ for $i \neq j$ (different color $\to$ zero)

Proof: Step 6 — Each R_i Is a Projection

From Step 5, the **diagonal blocks** give us:

$$V_0 U_0 = I_{r_0}, \quad V_1 U_1 = I_{r_1}, \quad \ldots, \quad V_k U_k = I_{r_k}$$

Now compute R_i^2. Remember $R_i = U_i V_i$:

$$R_i^2 = (U_i V_i)(U_i V_i) = U_i \underbrace{(V_i U_i)}_{=I_{r_i}} V_i = U_i V_i = R_i \quad \checkmark$$

Each R_i is **automatically** a projection!

Proof: Step 7 — Mutual Annihilation

From Step 5, the **off-diagonal blocks** give us: $V_i U_j = 0$ for $i \neq j$.

Now compute $R_i R_j$ for $i \neq j$:

$$R_i R_j = (U_i V_i)(U_j V_j) = U_i \underbrace{(V_i U_j)}_{=0 \text{ (off-diagonal!)}} V_j = 0 \quad \checkmark$$

The R_i are **automatically** mutually annihilating!

Finally, since $s = n$:

$$\text{rank}(R_0) + \text{rank}(R_1) + \cdots + \text{rank}(R_k) = n$$

$$\implies \text{rank}(R_1) + \cdots + \text{rank}(R_k) = n - \text{rank}(R_0) = \text{rank}(P) \quad \checkmark \qquad \square$$

Summary of the Proof

The entire proof flows from one chain:

$$P = R_1 + \cdots + R_k \xrightarrow{\text{add } R_0} I_n = R_0 + R_1 + \cdots + R_k$$
$$\xrightarrow{\text{factor}} UV = I_n \xrightarrow{s \leq n} s = n,\ VU = I_n$$

From the **block structure** of $VU = I_n$:

- **Diagonal blocks**: $V_i U_i = I_{r_i} \implies R_i^2 = R_i$ (projection!)
- **Off-diagonal blocks**: $V_i U_j = 0 \implies R_i R_j = 0$ (compatible!)

Key: The rank condition forces $s \leq n$, which forces $VU = I_n$, which forces **everything**.

Exercise: Verify the Rank Condition

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1,\ \text{rank}=1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

$\text{rank}(R_1) + \text{rank}(R_2) = 1 + 1 = 2 = \text{rank}(P)$ ✓

By the theorem: R_1, R_2 are automatically projections and $R_1 R_2 = 0$.

We get this **for free** — no need to multiply matrices!

Exercise: When the Rank Condition Fails

Same P, but a **different** decomposition:

$$P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{R_1,\ \text{rank}=2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_2,\ \text{rank}=1}$$

$\text{rank}(R_1) + \text{rank}(R_2) = 2 + 1 = 3 > 2 = \text{rank}(P)$

Rank condition **fails**. And indeed, R_1 is **not** a projection ($R_1{}^2 \neq R_1$).

Applications

Application 1: Cross-Filling Produces Compatible Projections

Corollary

Let P be a projection. Cross-fill $P = R_1 + \cdots + R_r$ into rank-1 pieces. Then $\{R_1, \ldots, R_r\}$ is a compatible family of rank-1 projections.

Proof: By construction, each R_i has rank 1, and:

$$\text{rank}(R_1) + \text{rank}(R_2) + \cdots + \text{rank}(R_r) = \underbrace{1 + 1 + \cdots + 1}_{r} = r = \text{rank}(P)$$

Apply the Projection Decomposition Theorem. $\square$

In Lecture 8, we proved this directly. The theorem provides a **second, more powerful proof** that generalizes far beyond cross-filling.

Application 2: Rank-1 Projections Have $r_i^T c_j = \delta_{ij}$

Cross-filling a projection P of rank r:

$$P = \underbrace{c_1 r_1^T}_{R_1} + \underbrace{c_2 r_2^T}_{R_2} + \cdots + \underbrace{c_r r_r^T}_{R_r}$$

Write as $P = UV$ where:

$$U = \begin{pmatrix} | & & | \\ c_1 & \cdots & c_r \\ | & & | \end{pmatrix}, \quad V = \begin{pmatrix} - & r_1^T & - \\ & \vdots & \\ - & r_r^T & - \end{pmatrix}$$

The theorem gives $VU = I_r$, which means:

$$r_i^T c_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Verify with Our Example

$$R_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 \end{pmatrix}$$

Check $r_i^T c_j = \delta_{ij}$:

$$r_1^T c_1 = \begin{pmatrix} -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 1 \checkmark \qquad r_1^T c_2 = \begin{pmatrix} -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 \checkmark$$

$$r_2^T c_1 = \begin{pmatrix} -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0 \checkmark \qquad r_2^T c_2 = \begin{pmatrix} -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 1 \checkmark$$

Application 3: Inner Cross-Filling — Motivation

Cross-filling decomposes P directly into rank-1 pieces.

But what if P has a **factored form**?

$$P = M_1 M_2 \cdots M_k$$

Idea: Instead of cross-filling the $n \times n$ matrix P, cross-fill a **smaller factor** inside P.

The Projection Decomposition Theorem will guarantee that the resulting pieces still form a compatible family!

Inner Cross-Filling: Theorem

Theorem (Inner Cross-Filling)

Let P be a projection. Suppose $P = M_1 M_2 \cdots M_k$ where $\text{rank}(P) = \text{rank}(M_i)$ for some factor M_i.

Cross-fill $M_i = Q_1 + Q_2 + \cdots + Q_s$ into rank-1 pieces. Define:

$$S_j = M_1 \cdots M_{i-1}\, Q_j\, M_{i+1} \cdots M_k$$

Then $P = S_1 + \cdots + S_s$ is a compatible family of projections.

Remark: We always have $\text{rank}(P) = \text{rank}(M_1 \cdots M_k) \leq \text{rank}(M_i)$ (rank of product $\leq$ rank of each factor). So "$\text{rank}(P) = \text{rank}(M_i)$" is asking that M_i is the **bottleneck** factor — no other factor squeezes the rank further.

Inner Cross-Filling: Why It Works

Clearly $P = S_1 + S_2 + \cdots + S_s$ (distribute the product over the sum).

For rank: each $\text{rank}(S_j) \leq \text{rank}(Q_j) = 1$, so:

$$\text{rank}(S_1) + \cdots + \text{rank}(S_s) \leq \underbrace{1 + \cdots + 1}_{s} = \text{rank}(M_i) = \text{rank}(P)$$

By the Projection Decomposition Theorem: each S_j is a projection and $S_j S_l = 0$ for $j \neq l$. $\square$

Key: We decompose an **inner factor** M_i rather than P itself. The outer factors are "carried along for the ride."

Inner Cross-Filling: Example

Let $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ (projection onto xy-plane). Factorize:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}}_{U} \cdot \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{I_2} \cdot \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{V}$$

$\mathrm{rank}(P) = 2 = \mathrm{rank}(I_2)$ ✓. Cross-fill I_2:

$$I_2 = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{Q_1} + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{Q_2}$$

Inner Cross-Filling: Example (continued)

$$S_1 = UQ_1V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$S_2 = UQ_2V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$S_1^2 = S_1$ ✓, $S_2^2 = S_2$ ✓, $S_1S_2 = 0$ ✓

We decomposed the 3×3 projection by cross-filling the **smaller** 2×2 identity!

Non-Trivial Example: Setup

Same projection $P = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$, but a **different** factorization:

$$\begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}}_{U} \cdot \underbrace{\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}}_{M} \cdot \underbrace{\begin{pmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}}_{V}$$

Here M is **not** the identity! But $\operatorname{rank}(P) = 2 = \operatorname{rank}(M)$ ✓.

Non-Trivial Example: Inner Cross-Fill M

Cross-fill $M = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ at pivot $(1,1)$:

$$Q_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$S_1 = UQ_1V = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S_2 = UQ_2V = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$S_1^2 = S_1$ ✓, $S_2^2 = S_2$ ✓, $S_1S_2 = 0$ ✓

Key: We cross-filled a 2×2 factor M (not P itself), and the theorem guarantees compatibility!

Inner Diagonal Cross-Filling for Orthogonal Projections

Combining Inner and Diagonal Cross-Filling

From Lecture 8: diagonal cross-filling decomposes an orthogonal projection P into rank-1 **orthogonal** projections.

From §4: inner cross-filling lets us decompose a **smaller factor** inside P.

Question: Can we combine these ideas?

If $P = UMU^T$ where M is a **small symmetric** matrix, can we diagonally cross-fill M and get orthogonal projections?

The Setup: $P = UMU^T$

Suppose an orthogonal projection has the form:

$$P = UMU^T$$

where:

- U is $n \times r$ with **full column rank** ($\text{rank}(U) = r$)
- M is $r \times r$ and **symmetric** ($M = M^T$)
- $\text{rank}(P) = \text{rank}(M)$

Key observation: M is a **much smaller** matrix than P.

M is $r \times r$ vs. P is $n \times n$. When $r \ll n$, this is a big difference!

Inner Diagonal Cross-Filling: Theorem

Theorem (Inner Diagonal Cross-Filling)

Let $P = UMU^T$ be an orthogonal projection and $\text{rank}(P) = \text{rank}(M)$.

Diagonally cross-fill M:

$$M = Q_1 + Q_2 + \cdots + Q_r$$

Define $R_j = UQ_jU^T$. Then:

1. $P = R_1 + R_2 + \cdots + R_r$
2. Each R_j is a rank-1 **orthogonal** projection
3. $\{R_1, \ldots, R_r\}$ is a compatible family

When Is $\text{rank}(P) = \text{rank}(M)$ Automatic?

If U is an $n \times r$ matrix with **full column rank** ($r \leq n$, "tall"), then:

$U\mathbf{x} \neq \mathbf{0}$ for every $\mathbf{x} \neq \mathbf{0}$ (full column rank means no non-trivial null space)

So $\text{rank}(UMU^T) = \text{rank}(M)$: the U and U^T factors **do not reduce rank**.

Practical rule: If U has full column rank, $\text{rank}(P) = \text{rank}(M)$ is **automatic** — no need to check!

Why It Works

Step 1: $R_1 + \cdots + R_r = UQ_1U^T + \cdots + UQ_rU^T = U(Q_1 + \cdots + Q_r)U^T = UMU^T = P$ ✓

Step 2 (rank): $\text{rank}(R_j) \leq \text{rank}(Q_j) = 1$, so:

$$\text{rank}(R_1) + \cdots + \text{rank}(R_r) \leq \underbrace{1 + \cdots + 1}_{r} = \text{rank}(M) = \text{rank}(P)$$

By PDT: each R_j is a projection and $R_jR_l = 0$. ✓

Step 3 (symmetry): Since Q_j is symmetric (diagonal pivot preserves symmetry):

$$R_j^T = (UQ_jU^T)^T = UQ_j^TU^T = UQ_jU^T = R_j \checkmark$$

Each R_j is a projection **and** symmetric $\implies$ orthogonal projection. $\square$

The Key Example: Orthogonal Projection Formula

The most important case: the orthogonal projection onto $\mathrm{Col}(B)$:

$$P = B(B^T B)^{-1} B^T$$

This has the form $P = UMU^T$ with:

- $U = B \quad (n \times r$, full column rank$)$
- $M = (B^T B)^{-1} \quad (r \times r$, symmetric $\checkmark)$
- $\mathrm{rank}(M) = r = \mathrm{rank}(P)$ $\checkmark$

Instead of diagonally cross-filling the $n \times n$ matrix P, we cross-fill the **small** $r \times r$ matrix $(B^T B)^{-1}$!

Efficiency Comparison

Method	**Size**	**Cost**
Direct diagonal cross-fill of P	$n \times n$	$O(n^2)$
Inner diagonal cross-fill of $(B^T B)^{-1}$	$r \times r$	$O(r^2)$

When $r \ll n$: inner diagonal cross-filling is **much faster**.

Example: B is 100×3 (3 basis vectors in $\mathbb{R}^{100}$).

Direct: cross-fill a 100×100 matrix.

Inner: cross-fill a 3×3 matrix!

Example: Start from a Concrete P

The same P from §3:

$$\underbrace{\frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}}_{P} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}}_{U} \cdot \underbrace{\frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}}_{M} \cdot \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}}_{U^T}$$

U has full column rank ✓, M is symmetric ✓.

Goal: Diagonally cross-fill the **small** 2×2 matrix M, then compute $R_j = UQ_jU^T$.

Example: Cross-Fill the Small M

$$M = \frac{1}{3}\begin{pmatrix} \mathbf{2} & -1 \\ -1 & 2 \end{pmatrix}$$

Diagonal pivot $(1,1)$, value $= \frac{2}{3}$:

$$Q_1 = \frac{1}{2/3} \cdot \frac{1}{3}\begin{pmatrix} \mathbf{2} \\ -1 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} \mathbf{2} & -1 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$$

$$Q_2 = M - Q_1 = \frac{1}{2}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$Q_1 = Q_1^T$ ✓, $\quad Q_2 = Q_2^T$ ✓ $\quad$ (diagonal pivots preserve symmetry!)

Example: Two Orthogonal Projections Emerge!

$$R_1 = UQ_1U^T = \frac{1}{6}\begin{pmatrix} 4 & -2 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$

$$R_2 = UQ_2U^T = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$R_1 = {R_1}^T$ ✓, $\quad \text{trace}(R_1) = 1$ ✓ $\quad \Longrightarrow$ **orthogonal** projection!

$R_2 = {R_2}^T$ ✓, $\quad \text{trace}(R_2) = 1$ ✓ $\quad \Longrightarrow$ **orthogonal** projection!

$R_1R_2 = 0$ ✓, $\quad R_1 + R_2 = P$ ✓

Same result as §3's direct cross-filling — but obtained from a 2×2 matrix instead of 3×3!

Preview: Spectral Decomposition

Compatible Families and Matrix Powers

Suppose $\{P_1, P_2, \ldots, P_n\}$ is a compatible family with $P_1 + \cdots + P_n = I$.

If $A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_n P_n$, then:

$$AB = (\lambda_1 P_1 + \cdots)(\mu_1 P_1 + \cdots) = \lambda_1 \mu_1 P_1 + \cdots + \lambda_n \mu_n P_n$$

because $P_i P_j = 0$ for $i \neq j$.

In particular:

$$A^k = \lambda_1^k P_1 + \lambda_2^k P_2 + \cdots + \lambda_n^k P_n$$

Example: Computing Matrix Powers

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 1 \cdot \underbrace{\frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}}_{P_1} + 3 \cdot \underbrace{\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{P_2}$$

$P_1^2 = P_1$ ✓, $P_2^2 = P_2$ ✓, $P_1 P_2 = 0$ ✓, $P_1 + P_2 = I_2$ ✓

Therefore:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^n = 1^n P_1 + 3^n P_2 = \frac{1}{2}\begin{pmatrix} 1+3^n & -1+3^n \\ -1+3^n & 1+3^n \end{pmatrix}$$

This is the main idea behind **spectral decomposition** (Chapter 5).

Summary

Key Results

1. **Compatible Families**: Projections $\{P_1, \ldots, P_k\}$ with $P_i P_j = 0$ — subsums are projections, vector decompositions are linearly independent.
2. **Projection Decomposition Theorem**: If $P = R_1 + \cdots + R_k$ with $\text{rank}(R_1) + \cdots + \text{rank}(R_k) \leq \text{rank}(P)$, then each R_i is a projection and $R_i R_j = 0$ — **automatically**.
3. **Cross-Filling**: Special case of the theorem — always produces compatible families.
4. **Inner Cross-Filling**: Decompose an inner factor of P — the outer factors carry along.
5. **Inner Diagonal Cross-Filling**: For $P = UMU^T$ with M symmetric, diagonally cross-fill the small $M \to$ orthogonal sub-projections.

Looking Ahead: Lecture 10

We have powerful tools for **decomposing** projections. But we still haven't answered:

Given a desired floor W and sunlight direction S, how do we construct P?

In Lecture 10:

- A projection is **uniquely determined** by $\text{Col}(P)$ and $\text{Null}(P)$
- Construction formula: $P = B(AB)^{-1}A$
- Orthogonal projection formula: $P = B(B^T B)^{-1}B^T$ (proof)
- Inner diagonal cross-filling of $(B^T B)^{-1} \to$ **orthogonal basis** for $\text{Col}(B)$